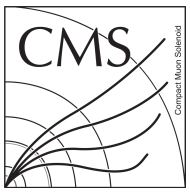
Internal review draft – 22 October 2026
No CMS DP number assigned



The Compact Muon Solenoid Experiment
CMS Performance Note



MCP-referenced timing calibration of CMS BTL

CMS

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Abstract

Several analyses of the 2025 data set benefit from an external time reference (t_0). During the testbeam period a Hamamatsu MCP-PMT was placed downstream of the setup and waveforms were digitized on a Lecroy scope. Since the active area ($\emptyset$=10mm) of the PMT only covered a couple BTL bars under test and the scope DAQ captured only 7.5k events per spill, we used a limited data set to calibrate Reference Bar timing relative to the MCP (which had intrinsic resolution of 7.5 picoseconds). The calibration functions are provided in this note.

Draft v1 for scientific discussion. Not approved for public release.
The following slides retain explicitly marked review items.

MCP-referenced timing calibration of CMS BTL

CMS

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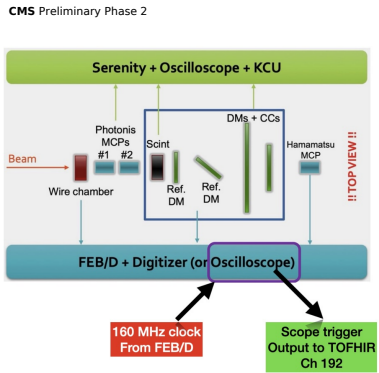
Discussion draft; not approved for public release.

Abstract

This study examines a timing-calibration procedure for CMS barrel timing layer (BTL) prototype scintillator bars using September 2025 beam-test data. A microchannel-plate photomultiplier (MCP-PMT), digitized by an oscilloscope, provides an external arrival-time reference. A trigger link identifies coincident events in the BTL readout and the oscilloscope. Segment-dependent clock alignment and locally weighted smoothing map the reference into the BTL time coordinate.

Quadratic functions of inverse charge describe the charge dependence of the measured arrival time for four silicon-photomultiplier channels on two reference bars. These functions provide an empirical time-walk calibration over the measured charge range. Clock-corrected profiles provide the reference for the companion dual-threshold study; archived fits are retained to document the change in calibration procedure. Timing-difference distributions illustrate the validation strategy. Their interpretation separates intrinsic reference-detector resolution from residual synchronization effects. This draft presents the results and the selection, uncertainty and reproducibility checks that remain before publication.

Beam-test setup and the reference signal



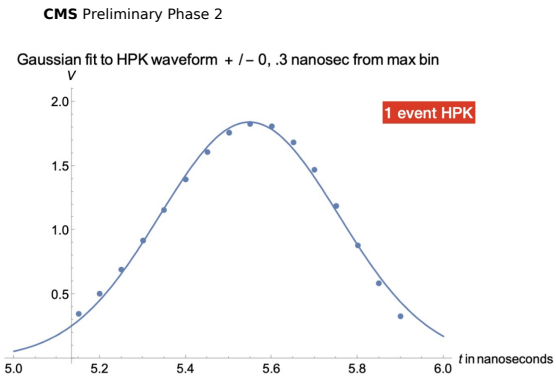
The September 2025 beam test used scintillator bars read out at both ends by silicon photomultipliers (SiPMs). A downstream MCP-PMT provides a fast reference for a restricted region of the bars.

The BTL front-end readout and oscilloscope use independent time coordinates. The scope trigger reaches TOFHIR channel 192; a separate 160 MHz clock link can also be recorded by the scope.

The reference-bar calibration uses the highlighted lower readout path.

Original setup illustration. Detector and readout context: Ref. [4]; the run-specific setup is described in this study.

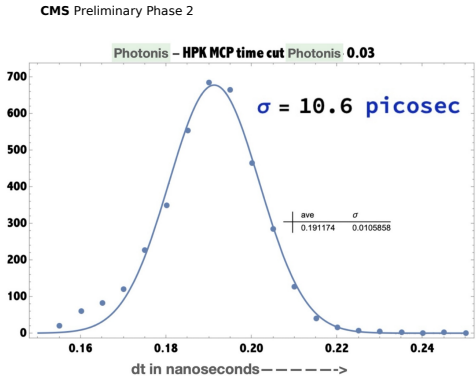
Reconstructing the MCP pulse time



The figure shows a Hamamatsu MCP waveform and its fitted pulse template. The fit estimates the pulse centroid rather than using the threshold-crossing time alone, reducing sensitivity to pulse amplitude.

The scope-trigger time identifies the event; the fitted peak supplies its relative reference time. Their difference is measured within the same oscilloscope waveform.

Intrinsic reference timing is a separate measurement

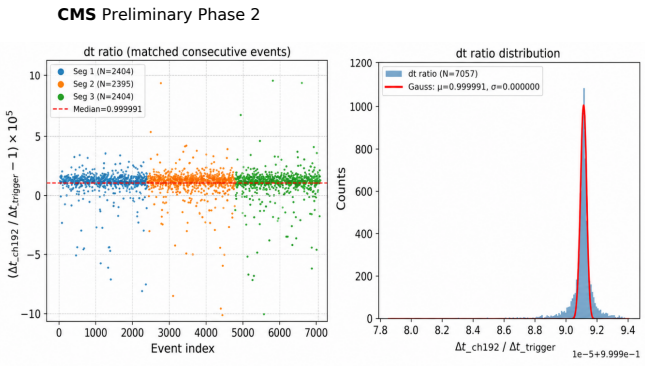


The distribution shows the difference between the fitted times of two MCP-PMTs recorded by the oscilloscope. Its measured width is 10.6 ps.

Assuming independent errors and a 7.5 ps PHOTONIS contribution gives $\sigma_{\mathrm{HPK}} \simeq \sqrt{10.6^2 - 7.5^2} = 7.5$ ps.

This scope-only estimate does not include synchronization to the BTL readout.

Matching the same particles between readouts



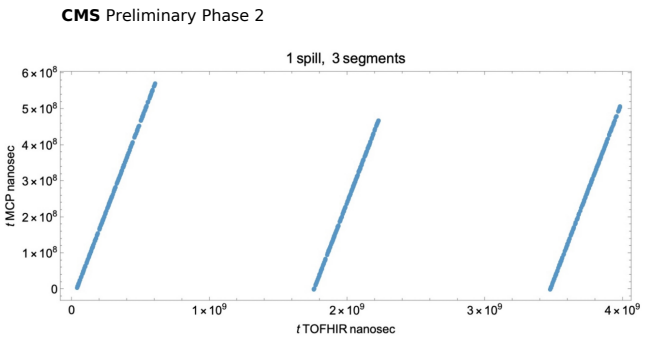
The matching procedure compares consecutive trigger intervals measured by channel 192 with those recorded by the scope. Correlated intervals identify candidate matches despite the independent clocks.

The interval-ratio distribution tests matching consistency. It is not a measurement of the picosecond resolution of the reconstructed particle time.

Only uniquely matched events enter the reference calibration.

GitHub: trigger-matching implementation. `dedup_triggers`: duplicate removal. `match_cumulative_time`: matching and tolerance.

Clock-rate alignment within acquisition segments

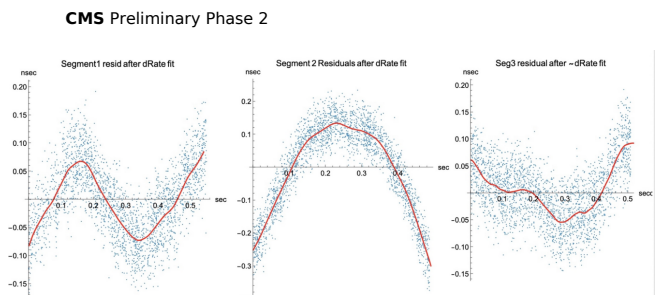


The figure compares matched scope-trigger times with TOFHIR trigger times. A nearly linear relation describes the overall clock-rate difference within an acquisition segment.

Both an offset and a slope are needed. Even a slope close to one matters over a long time interval. Clock resets require separate segment coordinates.

Archived clock-relation plot. It illustrates synchronization, not charge-dependent time walk.

Correcting smooth residual clock drift



After the linear clock-rate correction, the time residual varies within each of three segments. Points represent matched events; red curves show the smooth trend estimated by LOWESS, a locally weighted regression [8]. Adding the fitted trend to the linear prediction reconstructs the expected channel-192 trigger time. The reference difference subtracts this expected time once from the SiPM time.

Archived residuals in ns versus segment time in seconds. These are pre-LOWESS residuals, not final timing resolutions. GitHub: clock prediction; reference subtraction.

A single, explicit time convention

Let $g(u)$ map a scope-trigger timestamp u into the TOFHIR time coordinate, including the linear clock relation and the fitted smooth drift. Define the within-waveform difference

$$\delta_{\mathrm{MCP}} = t_{\mathrm{peak}}^{\mathrm{scope}} - t_{\mathrm{trigger}}^{\mathrm{scope}}.$$

The reference difference implemented in the calibration is

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t_{\mathrm{trigger}}^{\mathrm{scope}}) - \delta_{\mathrm{MCP}},$$

up to a fixed cable and geometry offset. The clock-rate correction for the short within-waveform interval is negligible and is omitted.

The charge-dependent component of Δt_i is the time walk. An arbitrary constant does not change a shape calibration.

GitHub implementation: per-file subtraction; combined-run subtraction. Both per-file and combined-run calculations subtract (`mcp_peak - mcp_trig`).

Reference bars and calibration selection

Upstream bar	SiPM channels	Role
7	133, 154	First reference pair
8	137, 150	Second reference pair

- Select coincident SiPM hits and a unique MCP-associated event, using channel 192 as the TOFHIR indicator.
- Require a successfully reconstructed MCP pulse and consistent peak-to-trigger timing. This is a waveform-quality cut, not a particle transit-time measurement inside the MCP.
- Use Q, the TOFHIR ADC charge proxy. It is not a deposited energy in MeV.
- Fit the profile of Δt_i versus $1/Q$ over the retained charge range.

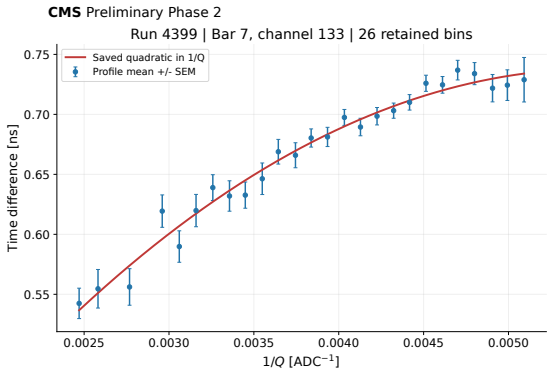
GitHub implementation:

Coincidence and walk-sample selection.

MCP mapping and optional cuts.

Peak-to-trigger timing selector.

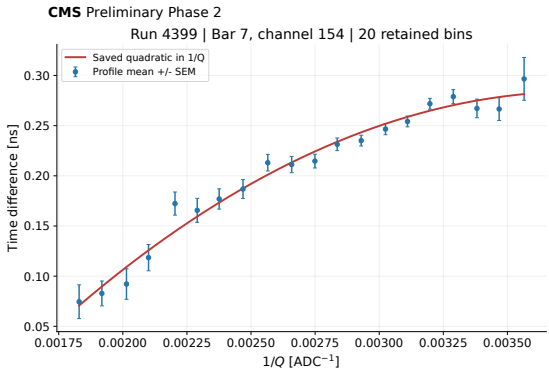
Archived reference: bar 7, channel 133



Run 4399 events are grouped in inverse-charge bins. Points show mean time differences with standard errors of the mean (SEM). The curve is the archived fit, $W(Q) = a + b/Q + c/Q^2$. There are 26 retained profile points. Times include the historical display offset. These archived means use measured channel-192 timing. An unweighted fit to the means reproduces the archived coefficients; SEM bars are not fit weights.

Archived profile and coefficients, displayed without refitting. The display convention includes a +26 ns shift.

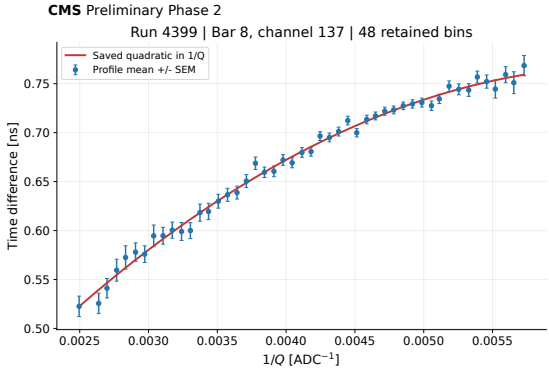
Archived reference: bar 7, channel 154



The opposite end of bar 7 is calibrated independently. Points show the mean time difference and SEM in 20 retained inverse-charge bins.
The red curve is the archived quadratic in $1/Q$, not a quadratic in Q. Channel-dependent coefficients allow for different pulse responses at the two bar ends. The archived fit is unweighted and uses measured channel-192 timing. The retained bin counts are channel dependent.

Run 4399 archived calibration, redrawn without refitting. Same +26 ns display convention as channel 133.

Archived reference: bar 8, channel 137

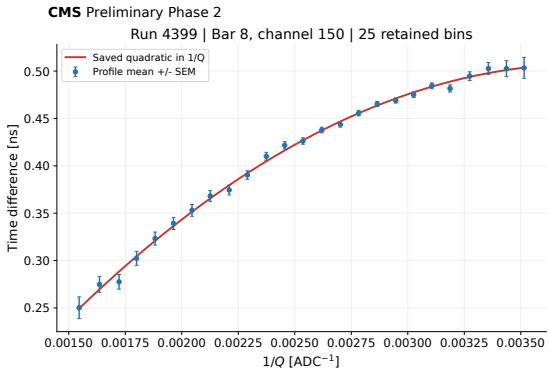


The channel-137 time difference increases as inverse charge increases: smaller signals cross a fixed threshold later.

Points are profile means with SEM; the red line is the archived quadratic fit to the profile. There are 48 retained points. The correction applies within the measured charge range; extrapolation is not validated. This unweighted result uses measured channel-192 timing and is retained for comparison with the clock-corrected reference shown next.

Run 4399 archived calibration. Coefficients and profile points are included with the draft.

Archived reference: bar 8, channel 150



The opposite end of bar 8 has its own charge-dependent delay. The 25 points show the mean timing difference and SEM; the curve is the archived quadratic fit to the profile.
The four curves demonstrate the calibration procedure. Each channel retains its original number of profile points and its original binning. This is the archived, unweighted fit using measured channel-192 timing.

Run 4399 archived calibration. Coefficient units are ns, ns ADC and ns ADC^2 for a, b and c.

Clock-corrected reference shared with Note 2

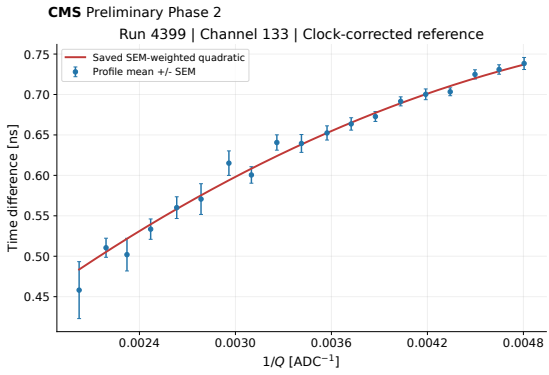
The following results use the expected trigger time $g(u)$ reconstructed from the segment-wise linear clock relation and LOWESS drift:

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t_{\mathrm{trigger}}^{\mathrm{scope}}) - (t_{\mathrm{peak}}^{\mathrm{scope}} - t_{\mathrm{trigger}}^{\mathrm{scope}}).$$

- Run 4399: exactly five TOFHIR hits, comprising the target upstream pair, downstream channels 234 and 243, and channel 192, with a unique MCP match.
- Require finite, positive charge and slope-related quality quantities; retain Q at or above the left half-maximum boundary of the spectrum peak.
- Form 20 uniform $1/Q$ bins; retain times within $5(1.4826\,\mathrm{MAD})$ of each bin's median, with at least five entries. Bars show the SEM after clipping.
- Fit $W(Q) = a + b/Q + c/Q^2$ with an objective weighted by $1/\mathrm{SEM}^2$. The next four figures use the same reference coefficients as Note 2.

The displayed $+26$ ns constant changes neither the shape nor the saved coefficients.

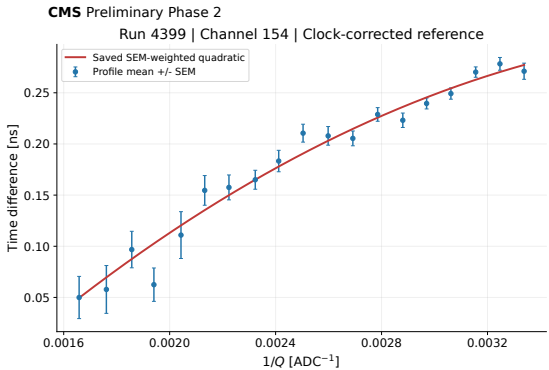
Clock-corrected reference: bar 7, channel 133



The time difference now uses the reconstructed trigger time rather than the measured channel-192 timestamp, suppressing the fitted clock drift.
Points show mean time differences with SEM in 19 retained inverse-charge bins.
The red curve is the SEM-weighted quadratic used in Note 2.
The curve is restricted to retained charge bins and the range between the outer profile means. Gaps from missing bins are left blank.

Run 4399; the reference used in Note 2, redrawn without refitting. A constant +26 ns shift is applied only for display.

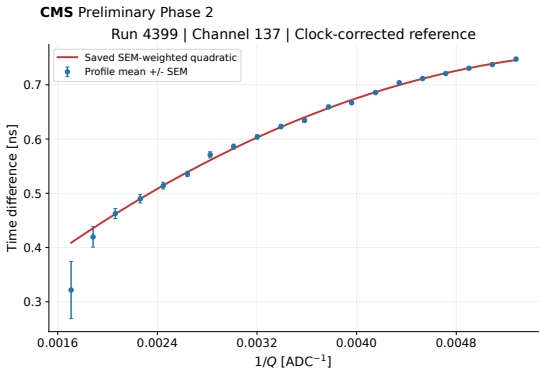
Clock-corrected reference: bar 7, channel 154



The same clock reconstruction and profiling procedure calibrates the opposite end of bar 7. Independent coefficients describe its charge-dependent timing response. The 19 points are retained-bin means; vertical bars show SEM after timing clipping. The red curve is the SEM-weighted quadratic in $1/Q$.

The reference used in Note 2. Coefficients and profile points accompany this note; the $+26$ ns shift is applied only for display.

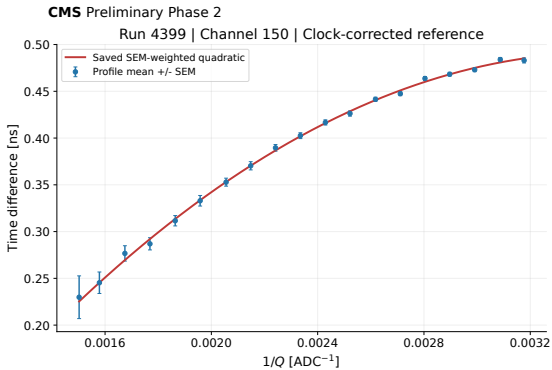
Clock-corrected reference: bar 8, channel 137



There are 20 retained inverse-charge bins for this channel. Smaller charge gives a larger timing delay, as expected for fixed-threshold readout of pulses with different amplitudes.
Points and bars are the mean and SEM after clipping. The saved quadratic is the MCP benchmark for the channel-137 inverse-slope comparison in Note 2.

Run 4399; expected-trigger timing; refitted after clock-quality selection. The +26 ns shift affects only the vertical display origin.

Clock-corrected reference: bar 8, channel 150



The channel-150 reference uses 19 retained profile bins and the same clock-corrected time definition as the other three channels.
Bars represent SEM, not the spread of individual event times. The red line is the SEM-weighted quadratic in inverse charge. These four results connect the MCP calibration in this note to the AWC shape comparison in Note 2.

Run 4399; coefficients are preserved in their original ns convention. Only the figure receives the constant +26 ns display shift.

Archived and clock-corrected shapes are close

The comparison uses both fitted functions in charge intervals populated in both samples, after removing a constant offset. No multiplicative scaling is fitted.

Channel	Shape RMS difference [ps]	Maximum absolute difference [ps]
133	3.20	10.63
154	4.29	9.68
137	1.33	3.22
150	4.91	10.47

The current reference identifies clock resets from finite timestamps and applies a clock-only quality requirement. The archived and current versions also differ in trigger reconstruction, selection, clipping and fit weights.

These are differences between calibration functions, not timing resolutions or a statistical-equivalence test. Note 2 uses the clock-corrected version.

Clock quality before timing validation

Segment boundaries are identified from consecutive finite scope timestamps, including resets across missing entries. Nonfinite trigger times have no valid mapping.

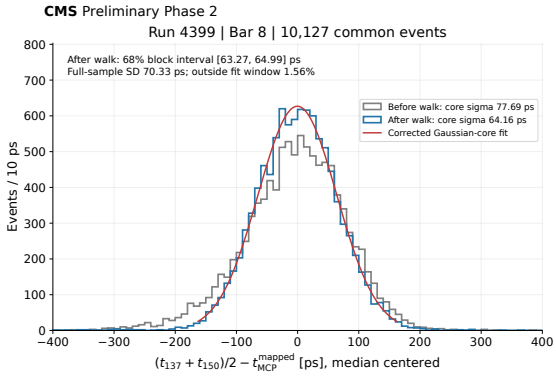
For each topology, file and segment, fit channel 192 against the scope trigger. Reject the whole group if the line-residual scale $1.4826\,\mathrm{MAD}$ exceeds 1 ns, before LOWESS. This rule uses clocks only, not SiPM arrival times.

Six bar-8 groups with finite trigger times fail this cut. Changing the threshold from 1 ns to 1 μs retains the same groups. Suspected one-event association offsets are excluded; MCP rows are not shifted to repair them.

Refit the channel walk curves after this selection. For the following timing comparison, both ends must pass the quality and lower-charge cuts. Each end must also lie within its populated reference bins and outer profile means. No additional selection is made on either timing-difference observable.

GitHub: clock-quality selection.

Timing validation against the MCP reference



For 10127 events, distributions show bar-average minus mapped MCP time before and after walk correction, with fixed clock mapping.

The core width changes from 77.69 to 64.16 ps. The corrected full-sample SD is 70.33 ps; 1.56% of events lie outside the fit window.

The conditional 16–84% interval is [63.27, 64.99] ps (400 file/segment-block bootstraps, fixed calibrations). Detector, MCP and residual clock effects contribute.

Truncated-normal fit: median $\pm 2.5(1.4826\,\mathrm{MAD})$. Median-centered display; full-sample SD; no segment-wise recentering.

What these timing widths establish

The reproduced distributions compare timing performance before and after correction under fixed selections and calibrations. They do not measure intrinsic bar resolution or isolate synchronization jitter. At fixed position, for end timing errors e_L, e_R,

$$\mathrm{Var}\Big(\frac{e_L + e_R}{2}\Big) = \frac{\mathrm{Var}(e_L - e_R)}{4} + \mathrm{Cov}(e_L, e_R).$$

Halving an end-difference width requires controlled position effects and negligible end-error covariance. Shared MCP and clock errors also prevent inferring detector covariance directly from the measured reference residuals.

Separately fitted core widths cannot be treated as variances of a common model to infer synchronization jitter by quadrature subtraction. This version reports same-event timing validation with conditional statistical intervals. Intrinsic bar resolution and synchronization-jitter extraction are outside its scope. Position, covariance, segment-offset and calibration systematics are not included in the quoted intervals.

Reference-calibration result and scope

- A trigger link enables calibration with common events across independent BTL and oscilloscope time coordinates.
- A linear clock relation plus a smooth residual correction defines the reference-time mapping.
- Four clock-corrected inverse-charge profiles and quadratic functions provide the reference used in Note 2. The archived fits remain as a comparison; profile points, SEM and both coefficient sets accompany this draft.
- Intrinsic MCP timing, reference-bar timing and cross-system synchronization are different performance quantities.
- Charge range and operating conditions limit transfer to other samples; a constant time offset is not a walk-shape difference.

For the same selected events, walk correction reduces the bar–MCP core width from 77.69 to 64.16 ps. This is a conditional timing-validation result, not an intrinsic-resolution measurement or a complete uncertainty budget.

References (1/2)

[1] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.

[2] Presentation by the Saclay group. `https://indico.cern.ch/event/1619706/`

[3] J. Bortfeldt et al., "Timing Performance of a Micro-Channel-Plate Photomultiplier Tube", Nucl. Instrum. Meth. A **960** (2020) 163592 [arXiv:1909.12604].

[4] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

References (2/2)

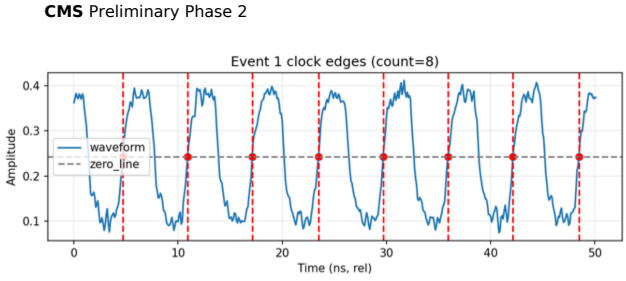
[5] S. White, Nov. 2025 BTL TB Analysis mtg presentation, https://indico.cern.ch/event/1614521/contributions/6803723/attachments/3178820/5922530/5vs10vs20GSa_s.pdf

[6] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[7] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[8] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

Backup: clock-phase reconstruction



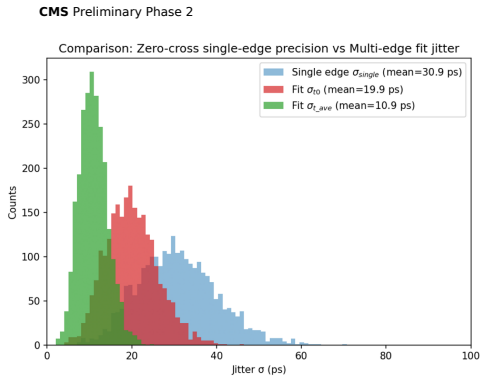
The reference clock is sampled by the oscilloscope. Several rising-edge crossings can be reconstructed within a waveform.

Fitting edge time against edge index estimates the clock phase from multiple cycles. Evaluating the fit near the mean edge index avoids unnecessary extrapolation.

This is an alternative synchronization diagnostic, distinct from the trigger-link method used for the archived reference curves.

Archived illustration. Correlated edge errors preclude assuming a $1/\sqrt{N}$ improvement in precision.

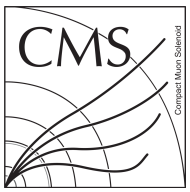
Backup: comparing clock estimators



The recorded MCP time is compared with three clock-phase estimators: a single edge, a regression intercept and the fitted time at the mean edge index. The distributions illustrate the dependence on the estimator. Their widths include all contributions present in this measurement and do not directly measure absolute clock accuracy. The underlying data, common selections and width definitions have not been established. This internal discussion figure is excluded from the publication candidate.

Archived clock-estimator diagnostic, retained for discussion.

Internal review draft – 22 October 2026
No CMS DP number assigned



The Compact Muon Solenoid Experiment
CMS Performance Note



Dual-threshold time-walk calibration for CMS BTL

CMS

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Abstract

With new high intrinsic precision timing detectors for the HL-LHC upgrades, the strategy for commissioning these detectors must differ from previous collider experience. Prior to LHC, collider experiments benefitted from a clear timing reference for use in calibration of the detectors. At low luminosity an interaction trigger was useful for rejecting background but also fulfilled this role. For the ATLAS and CMS upgrades, an interaction trigger would be superfluous since they are designed for 140-200 interactions per trigger event. Dual threshold timing enables a calibration strategy when a t_0 reference is not readily available. We extend an earlier bench test demonstration of this technique with the CMS Barrel Timing Layer(BTL) sensors and the TOFHIR readout ASIC to a 180 GeV pion beam exposure. These data confirm the earlier single coefficient correction for walk and the conclusion that walk calibration to the required $\sim 10\%$ precision is practical with simple data sets using only this system with beam or cosmic rays.

Draft v1 for scientific discussion. Not approved for public release.
The following slides retain explicitly marked review items.

Dual-threshold time-walk calibration for CMS BTL

CMS

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Discussion draft; not approved for public release.

Abstract

This study examines a charge-dependent time-walk model for CMS barrel timing layer (BTL) prototype channels using two discriminator thresholds. The current separation divided by the measured crossing-time separation provides a finite-interval pulse-slope proxy. A linear profile fit describes this proxy as a function of the measured charge, and an inverse-slope function models the timing correction with one amplitude parameter per channel.

The comparison uses slope measurements from run 4076 and independent microchannel-plate (MCP) reference calibrations from run 4399. The two runs contain different events. A fit determines a free constant offset and the correction amplitude only in charge intervals populated in both measurements. This is a comparison of calibration shapes, not an independent measurement of timing resolution.

The draft also examines channel-to-channel variations and an exploratory model that predicts the correction amplitude from internal pulse information. This model could enable operation without an event-by-event external time reference. Its transferability and full uncertainty remain to be established.

Two complementary runs, not the same events

Sample	Available information	Role in this study
Run 4076	Valid dual-threshold timing and charge	Measure the pulse-slope proxy $S(Q)$
Run 4399	MCP-associated reference timing and charge	Measure the empirical walk $W_{\mathrm{MCP}}(Q)$

The runs contain different events; no event-by-event matching is performed. The comparison transfers a channel-response model between runs.

The upstream and downstream modules each contain bars read out at two ends. Detector/readout context is given in Refs. [3, 7]. The channel-to-bar mapping determines the coincidence pairs; the same channel number is compared across the two runs.

The comparison assumes the same nominal detector setup in both runs, with MCP reference timing in run 4399 and dual-threshold timing in run 4076. Cross-run response differences are not included in the quoted uncertainties.

From two crossing times to a walk model

For thresholds $I_1 < I_2$ and their valid crossing times t_1, t_2, define

$$S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}], \qquad \overline{S}(Q) \simeq c_0 + c_1 Q.$$

S is a finite-interval rising-edge slope proxy. It is not the current difference ΔI itself and is not necessarily the instantaneous derivative at the first threshold.

$$W_Q(Q) = \frac{A}{c_0 + c_1 Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_Q(Q).$$

The amplitude A (AWC) has units of current and is constant for a channel and operating configuration. The correction varies event by event through Q.

Timing reconstruction and threshold scale

The run-4076 sample is file `23_e.root`. For coarse codes C_1, C_2,

$$d_c = (C_2 - C_1) \bmod 1024, \qquad t_2 = t_1 + (6250\,\mathrm{ps})d_c - f_1 + f_2.$$

Subtraction uses signed 64-bit coarse codes. The coarse-time term uses floating-point arithmetic to avoid the previous 16-bit multiplication overflow.

The fine-time lookup stores the mean of t_1 mod $6250\,\mathrm{ps}$ for each (channel, TAC, fine-code) combination, using all input hits. This lookup supplies f_1 and f_2, with entries for every selected hit in the 22 audited channels.

All selected hits have threshold register `step2=113621`, decoded as DAC codes 10 and 35. The implemented scale gives

$$\Delta I = (35 - 10) \times 0.3125\,\mu\mathrm{A} = 7.8125\,\mu\mathrm{A}, \qquad \frac{S}{\mu\mathrm{A/ns}} = 1000\,\frac{\Delta I/\mu\mathrm{A}}{(t_2 - t_1)/\mathrm{ps}}.$$

GitHub: timing and threshold reconstruction. This verifies the implemented conversion, not an independent calibration of the hardware thresholds.

Selection and profile construction

- Require one hit at each end of the target bar and no other hit in that module. The other module is unconstrained; channel 192 is ignored.
- Require finite Q, S, $\Delta I > 0$, $t_2 - t_1 > 0$ and $d_c = 0$. This selects the same-coarse-bin prompt population.
- Require $Q \geq Q_{\mathrm{left}}$, the left half-height boundary of the smoothed charge histogram. The Landau–Gaussian fit does not set this cut. There is no additional upper-charge cut or 100 ps minimum-rise cut.
- Form ten equal-width bins over the selected charge range. Plot mean S and hit SD; fit with SEM weights. Bins with undefined or negligible SEM are omitted.
- Bootstrap selected hits with fixed bin edges and refit 1000 times. Quote the 16th–84th percentiles.

All selected hits are used; there is no training or test split. GitHub: pair selection and profiling, with the newer optional minimum-rise cut disabled.

Reproducing the frozen run-4076 profiles

Reprocessing `23_e.root` reproduces the selected counts, ten-bin profiles and linear-fit coefficients for all 22 channels. The saved bootstrap coefficients reproduce the reported percentile intervals.

Channel	Target-pair hits	Prompt hits	Peak-right hits
137	25979	20789	19126

For channel 137, the histogram-derived boundary is $Q_{\mathrm{left}} = 184.9163$ ADC. Each profile point is placed at the mean measured Q in its bin, not at the geometric bin center.

The frozen selection retains rises below 100 ps in channels 130, 139, 151, 159, 236 and 238. These events remain in the sample. Their effect requires selection-sensitivity and channel-quality studies.

Channel 135 has one bin with a single entry. Its SEM is undefined, so only nine bins enter the linear fit. The audit changed neither the fits nor the plotted data.

Reference version shared with the MCP reference note [6]

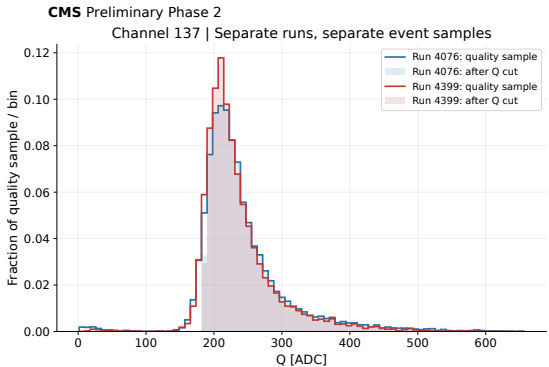
The run-4399 MCP functions were refitted on 22 October 2026. The updated procedure detects segment boundaries across missing trigger timestamps and excludes invalid clock groups. The MCP reference note documents the timing validation and remaining limitations.

For each topology, file and segment, the scaled MAD of the channel-192 versus scope-trigger line residuals must not exceed 1 ns before LOWESS [11]. The selection uses clock information, not SiPM timing residuals. Suspected incorrect event associations are excluded; MCP rows are not shifted.

All AWC-and-offset fits in this note use these updated references. The run-4076 slope fits, bootstrap replicas and measured-support matching definition are unchanged. The charge spectra use the updated reference sample.

The plotted AWC intervals hold the reference fixed. They exclude reference-fit uncertainty, clock-selection uncertainty and uncertainty in transfer between runs. The uncertainty table later in this note adds reference-fit statistics separately.

Charge spectra define the available comparison sample

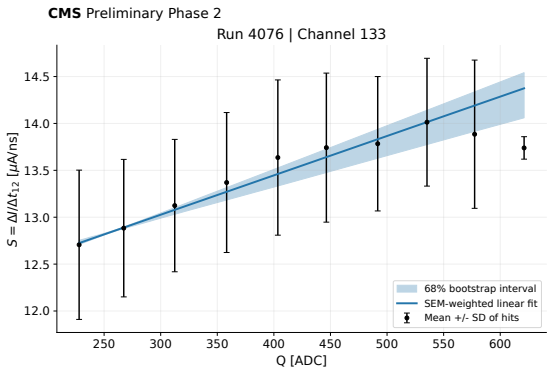


The spectra show the channel-137 charge distributions in the two independent runs. Lines show hits passing the quality selection; shaded spectra also require the charge-peak selection. Each run is normalized to its own quality-selected count.

The charge proxy is in ADC units, not MeV. The samples differ in population and acceptance, so the curve alignment uses only charge intervals populated in both runs.

Recovered histogram counts, not a new peak fit. Run-4076 profile selection starts at about 184.9 ADC for this channel.

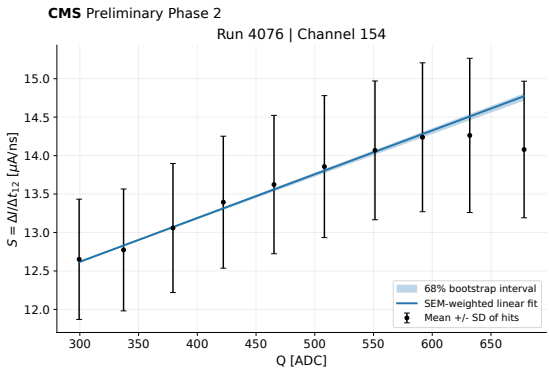
Channel 133: fitting the slope profile



Run 4076 channel-133 hits are grouped into ten charge bins. Points show mean S; bars show the SD of individual selected hits. The line is the SEM-weighted fit $c_0 + c_1 Q$.
The shaded interval is the central 68% bootstrap interval of the fitted mean curve, shown without enlargement.

SD is the hit standard deviation; SEM is the standard error of the mean. The band describes uncertainty on the fitted mean curve.

Channel 154: the opposite end of bar 7

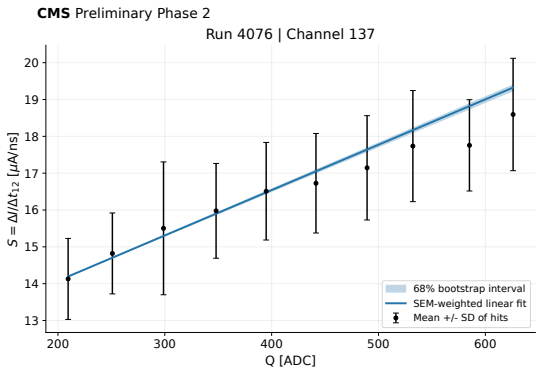


Channel 154 uses the same selection and profiling procedure. The mean slope increases with charge, but the response is calibrated independently at each bar end.

Bars show the hit SD. The line uses SEM weights. The band propagates the bootstrap coefficient replicas, including their correlation. Linearity is a working approximation, not a validated precision model. Model and selection dependence are excluded from the quoted errors; the separate sensitivity study is retained outside this note.

Run 4076, saved profile result, displayed without refitting.

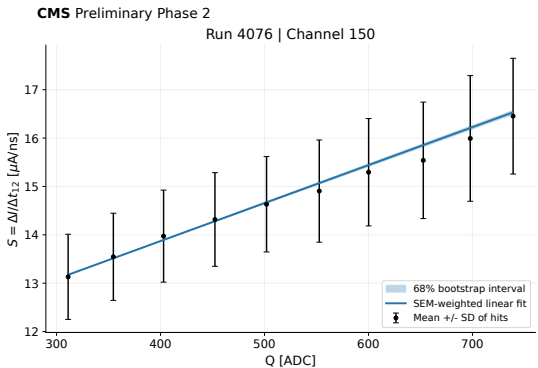
Channel 137: pulse slope and charge



The selected run-4076 sample contains 19 126 channel-137 hits. Ten profile points constrain a linear pulse-slope model over the observed charge range. The SD bars show event-to-event spread, while SEM weights determine the fit. The narrow shaded band is the conditional 68% bootstrap interval of the mean prediction, not a single-hit uncertainty.

The count and fit come from the saved profile results. Uncertainty bands are shown without enlargement.

Channel 150: the opposite end of bar 8



Channel 150 uses the same profiling procedure. The charge-dependent slope provides the denominator of the proposed walk function; a separate amplitude A determines the correction strength.

The line and shaded interval describe the fitted mean. Points and SD bars show the spread within the selected charge bins.

Run 4076, saved profile result. A constant AWC gives a charge-dependent timing correction.

Determine AWC using only jointly measured support

The run-4399 reference is a fitted quadratic in inverse charge,

$$W_{\mathrm{MCP}}(Q) = a + b/Q + c/Q^2.$$

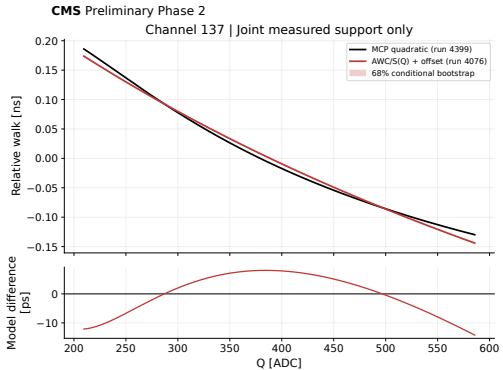
Let Ω be the intersection of the two outer profile-mean ranges, restricted to charge bins populated in both runs. Fit

$$(A, B) = \arg\min_{A,B} \sum_{Q_k \in \Omega} \left[\frac{A}{c_0 + c_1 Q_k} + B - W_{\mathrm{MCP}}(Q_k) \right]^2 .$$

The fit samples a uniform 3000-point charge grid, then removes intervals without data support. The offset B absorbs the arbitrary time zero.

This is uniform-charge shape matching, not an event-weighted likelihood. It uses neither extrapolated tails nor a single central anchor point. The uncertainty study below evaluates sensitivity to charge weighting and correlated MCP-fit uncertainty. Uniform Q is the nominal choice; it has not been established as uniquely optimal.

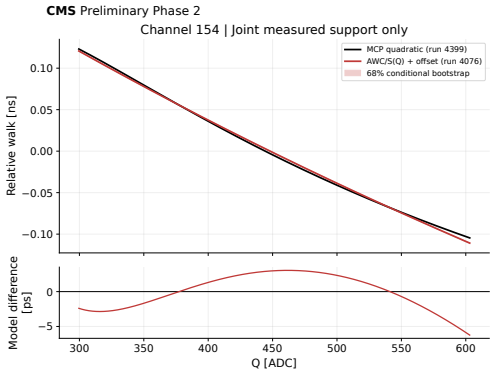
Channel 137: inverse-slope and MCP walk shapes



The black curve is the run-4399 MCP quadratic. The red curve is the run-4076 inverse-slope model after fitting AWC and a constant offset. The curves are shown only in charge intervals populated in both runs. Lower panel: run-4076 model minus MCP model. Its RMS shape mismatch is 6.80 ps on the supported uniform-Q grid, not an event timing resolution. The band varies the run-4076 slope fit while holding the MCP curve fixed. The uncertainty table later in this note also includes MCP-fit uncertainty.

AWC = 18.332 μA; conditional 16–84% interval [18.120, 18.607] μA. Both curves share one display offset.

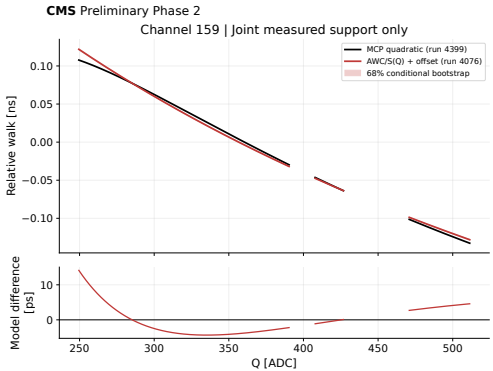
Channel 154: a second shape-comparison example



The same objective, restricted to the measured overlap, determines AWC and the offset for channel 154. Both plotted curves are calibrated models, not matched events from different runs. The lower panel shows the remaining shape mismatch, with no additional line subtracted. This measures model-shape closure, not corrected-event timing resolution.

Conditional bootstrap band: 4076 slope replicas, refitted AWC and offset, fixed 4399 quadratic.

Channel 159: missing support is not extrapolation

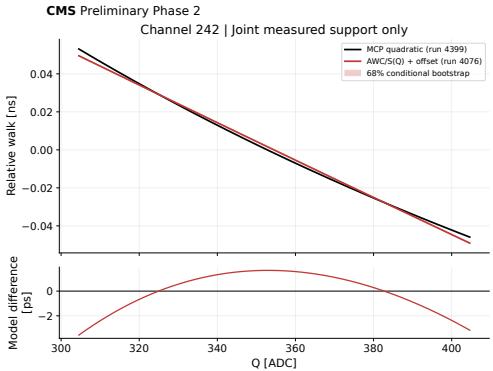


Only intervals with populated profile bins in both runs contribute to the alignment. Breaks in the displayed curves mark excluded charge intervals. They do not represent discontinuities in either analytic model.

The lower panel uses the same supported points. All displayed intervals participate in the common AWC-and-offset fit. It is a model difference, not an event-time distribution.

The comparison excludes 703 of the 3000 initial grid points for this channel. No line is drawn through those gaps.

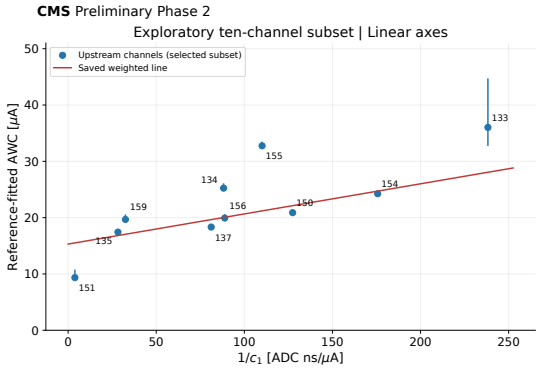
Downstream example: channel 242



This example applies the method to a downstream channel. In run 4399, upstream bar 8 supplies the coincidence pair alongside the downstream target pair and MCP indicator.
In run 4076, the target downstream pair is required, without fixing an upstream pair. The curves use only their common measured support. Downstream comparisons also assume the same nominal detector setup in both runs. Selection-dependent response differences remain unquantified.

The lower panel is a model difference, not a timing-resolution measurement. The other module is not used as an internal timing reference to determine AWC here.

A selected subset does not establish a universal law



The ten-channel upstream subset previously selected for discussion is shown on linear axes. Bars are slope-only statistical AWC bootstrap intervals; channel numbers identify the points.

The weighted straight-line fit has $\chi^2/\mathrm{ndf} = 538.6/8$ under its limited error model. It does not adequately describe the variation. The subset was selected interactively and is illustrative, not quality-selected or an unbiased population-level validation.

The fit treats x as fixed and uses half the 16–84% AWC span as its symmetric uncertainty. Correlations and reference errors are omitted.

Channel coverage and limitations

The explored set contains 14 upstream and 8 downstream channels, with 21 completed alignments. Only channel 238 lacks measured overlap. No additional quality cut is applied to this summary.

Channels	Observed limitation and treatment
238	No common measured charge range; no AWC estimate.
235, 244	Overlap spans only 3.10 and 16.99 ADC, respectively; one profile mean from each run lies within each overlap. Fits retained.
130, 236, 239	Slope-only AWC intervals include zero and span hundreds of μA. Fits retained as poorly constrained cases.
136, 245	Negative central AWC values are retained, not accepted as validated positive-amplitude calibrations.

The ten-channel illustration is unchanged: 133, 134, 135, 137, 150, 151, 154, 155, 156 and 159. This subset does not define a channel-selection criterion.

Table entries are diagnostic examples, not additional exclusions. Small curve mismatch over narrow support does not validate a calibration. The full per-channel support and interval table is saved with the figure data.

Uncertainties

Displayed quantity	Meaning
Profile bars	SD of selected hits in each charge bin; not the uncertainty on the mean.
Fit weights	SEM of each profile mean; not the SD.
Slope band	16–84% range from 1000 bootstrap linear fits with fixed bin edges.
AWC bars / walk band	Propagate the slope replicas through a new AWC-plus-offset alignment to the same fixed reference curve.

These definitions refer to the original displayed bands and bars. The next table additionally propagates the run-4399 quadratic covariance. Both prescriptions exclude uncertainty in response transfer between runs, clock and fine-time calibration, selection and model choices, and cross-channel correlations. Neither is a total calibration uncertainty or a timing-resolution estimate.

MCP-fit uncertainty and charge weighting

For each slope-bootstrap replica, propagate the correlated MCP coefficient covariance through the AWC-plus-offset fit. The Gaussian mixture gives the 16–84% interval; central AWC values and comparison curves are unchanged.

All table values are in μA. "Both runs" adds MCP-fit statistics, not cross-run or model systematics. The original plots retain slope-only statistical bands.

Channel	AWC	Slope only	Both runs	Max. weight shift
133	36.03	[32.73, 44.75]	[32.48, 45.72]	4.11
154	24.25	[23.92, 24.86]	[22.98, 25.84]	0.85
137	18.33	[18.12, 18.61]	[17.91, 18.80]	2.05
150	20.89	[20.65, 21.15]	[20.41, 21.38]	1.13

Maximum weighting shift: largest $|\Delta\mathrm{AWC}|$ using uniform $1/Q$ or either run's charge population instead of uniform Q. The overlap and free offset are unchanged. These shifts are not statistical errors and are not added in quadrature.

All 21 overlap channels were checked; channel 238 has no overlap. A shared reference-time offset affects only the fitted offset, not AWC. These are conditional statistical intervals, with weighting sensitivity reported separately, not total calibration uncertainties.

Result and next validation step

- A dual-threshold measurement provides a charge-dependent slope proxy; an inverse-slope function can be aligned to independent MCP walk curves.
- Restricting alignment to jointly measured charge support avoids determining AWC from extrapolation.
- Reference-calibrated AWC varies between channels. A straight line against $1/c_1$ is insufficient for the explored subset.
- The present calibration still uses the MCP to determine AWC. It does not establish reference-free AWC extraction by itself.
- Before use without MCP, an internal predictor must be tested on held-out bars and configurations, including uncertainties in transfer and in the target AWC values.

The following backup explores such a predictor; it is not a final performance claim or a demonstration with cosmic rays.

Backup: an internal predictor for channel AWC

For $W = A/S$ and $S = c_0 + c_1 Q$,

$$-\frac{dW}{d\ln Q} = \frac{A\, c_1 Q}{S(Q)^2}.$$

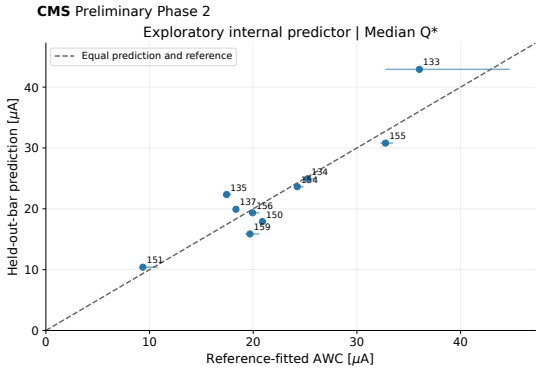
If this local walk derivative is approximately shared between channels at a prescribed typical charge Q_*, a candidate model is

$$A_{\mathrm{pred}} = K\, \frac{S(Q_*)^2}{c_1 Q_*}.$$

Here Q_* is the median charge of the run-4076 quality- and peak-right-selected sample. It is computed once per channel; A_{pred} remains constant when correcting individual events.

The common K is learned from MCP-calibrated training channels. Predictions for new channels use internal measurements, but model development uses the reference. The shared derivative and choice of Q_* remain hypotheses requiring independent validation across operating conditions, without tuning to the test results.

Backup: predict a bar excluded from the model fit



Each point compares reference-fitted AWC with a prediction made after excluding that channel's entire bar from the fit of K. Ten upstream channels span six held-out bar groups.
The RMS prediction error is 3.22 μA; the RMS fractional error is 14.1%.
Horizontal intervals are conditional reference-AWC intervals, not predictive intervals. This small selected sample leaves input–target correlations and full prediction uncertainties unpropagated. It does not demonstrate reference-free calibration.

Exploratory median-Q_* candidate. The dashed line indicates equality, not a fitted result. A general 10% precision claim is not established.

References (1/2)

[1] "Performance of the ALICE Time-Of-Flight detector at the LHC", Eur. Phys. J. Plus (2013) 128: 44

[2] "PID performance of the ALICE-TOF detector at Run 2", N. Jacazio in Proceedings of LHCP2018, https://pos.sissa.it/321/232/.

[3] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[4] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[5] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[6] S. White and Licheng J. Zhang, "September 2025 BTL Testbeam Reference Bar Walk Calibration from Hamamatsu MCP-PMT ",cms note.

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[7] E. Albuquerque et al., "TOFHIR2: The readout ASIC of the CMS Barrel MIP Timing Detector", arxiv:2404.01208

[8] J.E. Grund, SLAC-PUB-2416 (Oct. 1979).

[9] A.R. Frolov, T.V. Oslopova, Yu.N. Pestov, "Double threshold discriminator for timing measurements", Nuclear Instruments and Methods Volume 356, Issues 2/3 (1995) `https://doi.org/10.1016/0168-9002(94)01239-3`.

[10] This will arise in LGAD that are not thinned to remove the effect (J. Vavra, private communication). See also PICOSEC, "Fast Timing with MicroPattern Gas Detectors", where charge in the electron peak is captured- ignoring the ion tail..

[11] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

[12] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.